KITTI Splat

EC327 Software Engineering Final Project | Spring 2026

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1. Introduction & Project Goals

Overview

KITTI Splat is a high-performance 3D reconstruction pipeline that utilizes **3D Gaussian Splatting (3DGS)** to generate "digital twins" of urban environments from sparse autonomous driving data. By fusing RGB camera imagery with Velodyne LiDAR geometry and OXTS GPS trajectories, the project transforms a simple car drive into a navigable, photorealistic 3D environment.

Overall Goal

- Create a realistic, 3D scene of a chunk of KITTI dataset residential 0064.

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2. How It Works: Intuition & Architecture

The Intuition: "Painting with 3D Stickers"

Imagine trying to recreate a street by taking a photograph and then placing thousands of tiny, translucent, semi-transparent "stickers" (Gaussians) in 3D space to match that photo.

1. **LiDAR** provides the initial "skeleton" of where these stickers should go.
2. **GPS** tells us how the car moved so we can align stickers from different frames.
3. **Optimization** then adjusts the size, color, and transparency of these stickers until the 3D model looks exactly like the original video from any angle.

Fig 1. Example of training a section of the splat based on Image

Fig 2. Example of the final splat from realistic viewing angle

The Technical Workflow

The pipeline is divided into three distinct stages:

1. **Data Pre-processing:**
   The system reads raw LiDAR .bin files and transforms them into a global point cloud using the OXTS trajectory. It then projects these points into the camera's 2D frame using extrinsic and intrinsic matrices to ensure pixel-perfect alignment.
2. **Semantic Masking (SegFormer):**
   Because the sky is "at infinity," it confuses the 3DGS optimizer, leading to "sky spears" (geometry that stretches to the horizon). We use a pre-trained **SegFormer** model to identify sky pixels and mask them out, forcing the model to focus 100% of its learning capacity on the street and objects.
3. **Constrained Optimization:**
   We implement a custom **Densifier** that enforces hard physical limits on the model's growth, ensuring the model stays grounded and realistic.

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3. Design Decisions & Tradeoffs

Decision	Explanation	Tradeoff
3DGS vs. NeRF	Chose 3D Gaussian Splatting over Neural Radiance Fields because more interested in 3DGS.	**3DGS** offers near-instant real-time rendering speeds but requires more complex geometric initialization than NeRF.

Specific Scene Usage	Chose Residential_0064 as it was a smaller scene that could be easily stored.	Has many scenes that include the sky which gives us troubles with “sky spears.”
Only using a chunk of the scene	Only used a chunk of the scene, not the whole thing.	It gives us lots of unoptimized splats (gray blobs), but allows us to avoid memory issues while training (especially useful for Colab).

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4. Team Contributions: Who Did What?

- I was a solo group so I did everything :D

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5. Why This is "Cool" & Worthy of a Project

The "Cool" Factor

Most 3D models are built by hand in software like Blender. This project does the opposite: it **extracts reality**. By just driving a car down the street, we can generate a 3D world that you can physically "walk" through in a browser. It is the bridge between a flat video and a truly immersive Digital Twin. In addition, this is an unbounded world scene. Instead of just splatting an object, we have to deal with the sky, requiring us to pull out some fun tech (Segmentation Model) to deal with it.

Project Scope

This is not a standard homework assignment as it dives into the realms of many different problems in Computer Vision and Video processing. This project includes:

1. **3D Math:** Understanding 6-DOF coordinate transformations.
2. **State-of-the-Art AI:** Integrating 2024-standard research (3DGS and SegFormer).
3. **System Design:** Managing gigabytes of sparse data across cloud environments and optimizing for GPU memory constraints.

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6. Final Summary

KITTI Splat demonstrates that with a combination of LiDAR geometric priors and Gaussian Splatting optimization, we can create high-fidelity 3D reconstructions that are solid baselines to trying to create 3D models used in autonomous vehicle simulation. The final output - a clean, navigable .ply file - gives us a really awesome model of a scene that can be really hard to recreate with the given data. Overall, although not photo-realistic, I’m pretty proud of the project and what I was able to do given the time frame and other extending circumstances.